P.Vishwa Nandan(192473015)

4. Sales Performance Analysis

```
import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)
increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100

print("Total Sales:", total_sales)
print("Percentage Increase:", increase, "%")
```

Output:

```
Total Sales: 55000
Percentage Increase: 80.0 %
```

5. Fuel Efficiency Analysis

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40])

average = np.mean(fuel_efficiency)
improvement = ((fuel_efficiency[1] - fuel_efficiency[0]) /
        fuel_efficiency[0]) * 100

print("Average Fuel Efficiency:", average, "mpg")
print("Percentage Improvement:", improvement, "%")
```

Output:

```
Average Fuel Efficiency: 32.5 mpg
Percentage Improvement: 20.0 %
```

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6. Grocery Purchase Cost Calculation

```
prices = [100, 200, 50]

quantities = [2, 1, 3]


discount_rate = 10

tax_rate = 5


subtotal = 0


for i in range(len(prices)):
    subtotal = subtotal + prices[i] * quantities[i]


discount = subtotal * discount_rate / 100

discounted_amount = subtotal - discount

tax = discounted_amount * tax_rate / 100

total_cost = discounted_amount + tax


print("Subtotal:", subtotal)

print("Discount:", discount)

print("Tax:", tax)

print("Total Cost:", total_cost)
```

Output:

```
Subtotal: 550
Discount: 55.0
Tax: 24.75
Total Cost: 519.75
```